

Graded potentials

Name _____ Date _____

Use this sheet while you work through the graded potentials walkthrough. It has 12 predictions.

1. When a step asks you to predict, write or draw your prediction in the top box, in pencil or your first color. Type it into the worksheet beside the walkthrough too. Next stays locked until you press Show me.
2. Press Show me. In the bottom box, in a second color, write what actually happened and whether you were right. If you were not, write what you would change.
3. When you finish the walkthrough, close it and do Part 2 from memory. Then reopen it and correct Part 2 in your second color. What you had to fix is what to study.

Part 1. Your predictions

PREDICTION 1 OF 12 · A CHEMICAL OPENS A FEW DOORS

Sodium flows in through the open channels. Does the tracing move up or down, and why?

My prediction, before Show me

What happened, and was I right?

PREDICTION 2 OF 12 · PUSH HARDER

What happens to the size of the change in voltage as the stimulus gets stronger?

My prediction, before Show me

What happened, and was I right?

PREDICTION 3 OF 12 · A DOOR THAT GOES THE OTHER WAY

Potassium leaves through the open channels. Which way does the tracing move this time?

My prediction, before Show me

What happened, and was I right?

PREDICTION 4 OF 12 · CHARGE SPREADS BOTH WAYS

Where does the positive charge go after it enters the dendrite?

My prediction, before Show me

What happened, and was I right?

PREDICTION 5 OF 12 · MEASURE ALONG THE WAY

Will all four meters show the same change in voltage? If not, which will show the most?

My prediction, before Show me

What happened, and was I right?

PREDICTION 6 OF 12 · A LEAKY HOSE

What happens to some of the positive charge as it travels past those open leak channels?

My prediction, before Show me

What happened, and was I right?

PREDICTION 7 OF 12 · A WIDER DENDRITE

Will a graded potential travel farther or a shorter distance before it fades in the wider dendrite? Which of the two losses changes?

My prediction, before Show me

What happened, and was I right?

PREDICTION 8 OF 12 · ONE ARRIVES

Rest is -70 mV and threshold is about -55 mV. Does the trigger zone fire?

My prediction, before Show me

What happened, and was I right?

PREDICTION 9 OF 12 · TWO ARRIVE TOGETHER

Does the trigger zone fire now? Show the arithmetic.

My prediction, before Show me

What happened, and was I right?

PREDICTION 10 OF 12 · ADD A BRAKE

Does the trigger zone still fire?

My prediction, before Show me

What happened, and was I right?

PREDICTION 11 OF 12 · PATIENT: THE PUFFERFISH MEAL

Why do his graded potentials still happen, and what signal is missing?

My prediction, before Show me

What happened, and was I right?

PREDICTION 12 OF 12 · THE WHOLE STORY, WITH THE SCIENCE NAMES

Without scrolling back, explain why a graded potential that is 20 mV at the tip of a dendrite can fail to fire the neuron, and name two ways it could still reach threshold.

My prediction, before Show me

What happened, and was I right?

Part 2. Draw it from memory

Graded potentials

1. Graded potentials of different sizes

Draw a voltage tracing: membrane voltage in mV up the side, time along the bottom. Mark rest at -70 mV and threshold at about -55 mV. On it, draw the graded potential from a weak, a medium and a strong stimulus, and one that dips below rest. Label each one with what the open channels let in or out.

2. Why it fades as it spreads

Draw a dendrite as a long tube with the stimulus at one end. Show which way the charge spreads, draw four meters along it, and write the size of the change at each one. Label current leak and cytoplasmic resistance where each one happens.

Then answer: if the dendrite were twice as wide, what would change and what would stay the same?

3. Below threshold and above it

Draw two tracings at the trigger zone, each with rest and threshold marked. In the first, one graded potential arrives and fades. In the second, two arrive at the same time and add up to threshold. Write the voltage each one reaches.

One arrives

Two arrive together

4. Read a patient

A man who ate badly prepared pufferfish still makes small graded potentials, but none of his neurons fires an action potential, even when a strong stimulus takes the membrane past threshold. Explain which channels the toxin blocks, which channels it leaves alone, and why that explains what his recordings show.